

```
#include <stdio.h>
```

```
#define MAX 10
```

```
// Function to input a matrix
```

```
void inputMatrix(int matrix[MAX][MAX], int rows, int cols) {  
    int i, j;  
  
    for (i = 0; i < rows; i++) {  
        for (j = 0; j < cols; j++) {  
            scanf("%d", &matrix[i][j]);  
        }  
    }  
}
```

```
// Function to display a matrix
```

```
void displayMatrix(int matrix[MAX][MAX], int rows, int cols) {  
    int i, j;  
  
    for (i = 0; i < rows; i++) {  
        for (j = 0; j < cols; j++) {  
            printf("%d\t", matrix[i][j]);  
        }  
        printf("\n");  
    }  
}
```

```
// Function for matrix addition
```

```
void addMatrix(int A[MAX][MAX], int B[MAX][MAX], int result[MAX][MAX],  
               int rows, int cols) {  
    int i, j;  
  
    for (i = 0; i < rows; i++) {  
        for (j = 0; j < cols; j++) {  
            result[i][j] = A[i][j] + B[i][j];  
        }  
    }  
}
```

```
// Function for matrix multiplication
```

```
void multiplyMatrix(int A[MAX][MAX], int B[MAX][MAX],  
                   int result[MAX][MAX],  
                   int r1, int c1, int c2) {  
    int i, j, k;  
  
    for (i = 0; i < r1; i++) {  
        for (j = 0; j < c2; j++) {  
            result[i][j] = 0;  
  
            for (k = 0; k < c1; k++) {  
                result[i][j] += A[i][k] * B[k][j];  
            }  
        }  
    }  
}
```

```

// Function to find transpose
void transposeMatrix(int matrix[MAX][MAX],
                    int transpose[MAX][MAX],
                    int rows, int cols) {
    int i, j;

    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            transpose[j][i] = matrix[i][j];
        }
    }
}

int main() {
    int A[MAX][MAX], B[MAX][MAX];
    int sum[MAX][MAX], product[MAX][MAX], transpose[MAX][MAX];
    int r1, c1, r2, c2;

    // Input first matrix
    printf("Enter rows and columns of Matrix A: ");
    scanf("%d %d", &r1, &c1);

    printf("Enter elements of Matrix A:\n");
    inputMatrix(A, r1, c1);

    // Input second matrix
    printf("Enter rows and columns of Matrix B: ");
    scanf("%d %d", &r2, &c2);

    printf("Enter elements of Matrix B:\n");
    inputMatrix(B, r2, c2);

    // Matrix Addition
    if (r1 == r2 && c1 == c2) {
        addMatrix(A, B, sum, r1, c1);

        printf("\nMatrix Addition:\n");
        displayMatrix(sum, r1, c1);
    } else {
        printf("\nMatrix Addition not possible.\n");
    }

    // Matrix Multiplication
    if (c1 == r2) {
        multiplyMatrix(A, B, product, r1, c1, c2);

        printf("\nMatrix Multiplication:\n");
        displayMatrix(product, r1, c2);
    } else {
        printf("\nMatrix Multiplication not possible.\n");
    }
}

```

```
// Transpose of Matrix A
transposeMatrix(A, transpose, r1, c1);

printf("\nTranspose of Matrix A:\n");
displayMatrix(transpose, c1, r1);

return 0;
}
```

Enter rows and columns of Matrix A: 2 2

Enter elements of Matrix A:

1 2

3 4

Enter rows and columns of Matrix B: 2 2

Enter elements of Matrix B:

5 6

7 8

Matrix Addition:

6 8

10 12

Matrix Multiplication:

19 22

43 50

Transpose of Matrix A:

1 3

2 4